FASER tracking and alignment

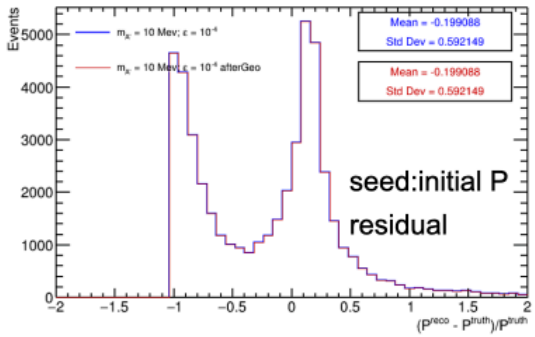
Xiaocong Ai, Nov 11, 2026

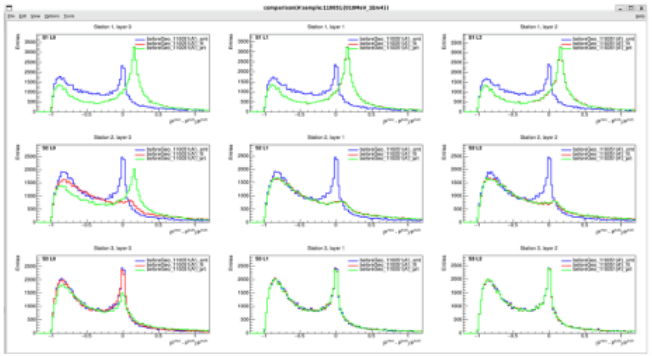
FASER alignment issue

- Four-station alignment with IFT works, but provides unphysical alignment parameters, i.e. alignment parameters are obviously larger than expected
  - Need to investigate how to constrain some alignment freedom and what is the best order of station/layer/module-level alignment using MC

FASER tracking issue

- Biased momentum distribution (even after geometry rotation)
  - Need to investigate the reason for the bias
    - Already see this in initial momentum estimated from seeds





CKF_perLayerPResidual{beforeGeo}:

blue line： smt state

green line: prt state

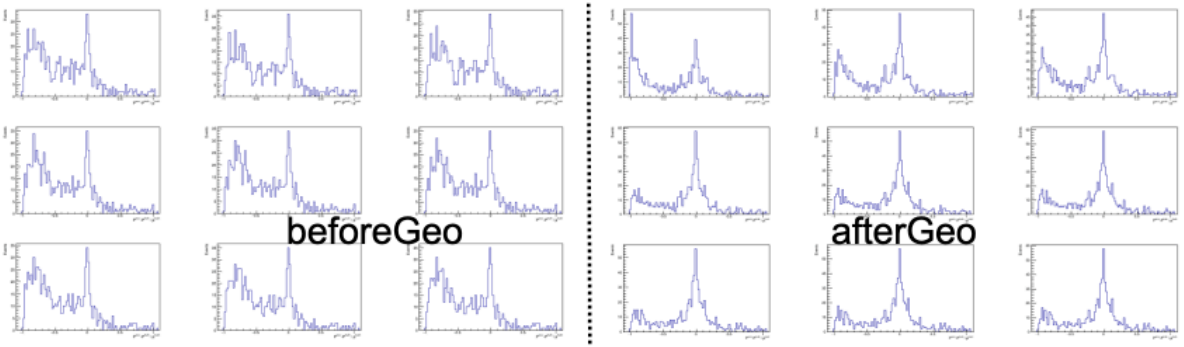
red line: flt state

    - but it can happen that good momentum does not necessarily result in good momentum after CKF, why?

seed limit:

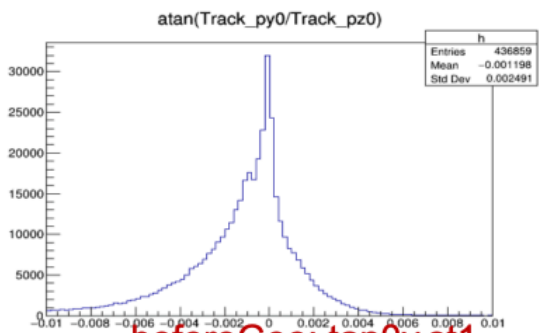
- std::abs(residualP) < 0.1
- the same Q sign
- std::abs(residualTheta)<0.0001

see the CKF →

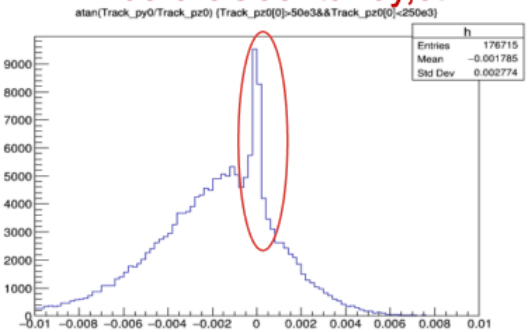


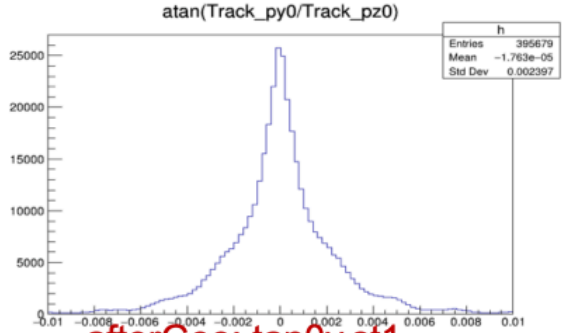
FASER tracking issue

- Need to confirm there is no unphysical peak after geometry rotation （see slides)

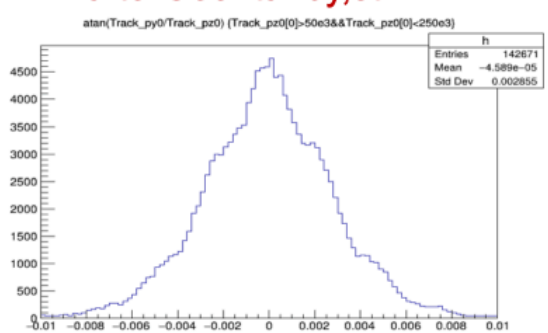


beforeGeo: tanθy,st1





afterGeo: tanθy,st1



The sharp peak near tan θy, st1 = 0 is more prominent, which is improved after geometry rotation.

- Higher charge mis-identification rate of mu- than mu+ (see slides)